Ruilong Li

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I am a currently research scientist at NVIDIA Spatial Intelligence Lab. I received my PhD from UC Berkeley, where I was advised by Prof. Angjoo Kanazawa in BAIR. My current research interests widely spread in Computer Vision & Graphics & Machine Learning, especially on the tasks of generation and reconstruction in the 3D world via information from 2D images.

EDUCATION

Ph.D - University of California, Berkeley — **Aug. 2021 – May. 2025**
Computer Science — *Berkeley, USA*

M.Sc - University of Southern California — **Aug. 2019 – May 2021**
Computer Science — *Los Angeles, USA*

M.Eng - Tsinghua University — **Aug. 2016 – May 2019**
Computer Science and Technology — *Beijing, China*

B.Sc - Tsinghua University — **Aug. 2012 – May 2016**
Mathematics and Physics — *Beijing, China*

RESEARCH POSITIONS

Research Scientist, NVIDIA Spatial Intelligence Lab — **April 2025 - Present**
Working with Zan Gojcic and Sanja Fidler — *Santa Clara, USA*

Research Intern, NVIDIA Spatial Intelligence Lab — **May 2023 – Dec 2024**
Working with Francis Williams and Sanja Fidler — *Santa Clara, USA*

Research Intern, Meta Reality Lab — **May 2021 – Dec 2021**
Working with Christoph Lassner — *Sausalito, USA*

Research Intern, Google Research — **May 2020 – Jan. 2021**
Working with Shan Yang and Angjoo Kanazawa — *Mountain View, USA*

Research Assistant, USC Institute for Creative Technologies — **Aug 2019 – May 2021**
Working with Hao Li — *Los Angeles, USA*

ACADEMIC SERVICES

Reviewer: CVPR, ICCV, ECCV, ACCV, WACV, 3DV, EuroGraphics, Siggraph Asia, ToG, TPAMI, TIP

OPEN SOURCE LIBRARIES

gsplat[link], nerfacc[link], nerfstudio[link]

PUBLICATIONS

- "Cameras as Relative Positional Encoding," *Advances in Neural Information Processing Systems (NeurIPS)*, 2025, **Ruilong Li***, Brent Yi*, Junchen Liu*, Hang Gao, Yi Ma, Angjoo Kanazawa.
- "gsplat: An open-source library for Gaussian splatting," *Journal of Machine Learning Research* (JMLR), 2025, Vickie Ye*, **Ruilong Li***, Justin Kerr, Matias Turkulainen, Brent Yi, Zhuoyang Pan, Otto Seiskari, et al.
- "NeRF-XL: Scaling NeRFs with Multiple GPUs," *European Conference on Computer Vision (ECCV)*, 2024, **Ruilong Li**, Sanja Fidler, Angjoo Kanazawa, Francis Williams.
- "fvdb: A deep-learning framework for sparse, large scale, and high performance spatial intelligence," *ACM Transactions on Graphics (TOG)*, 2024, Francis Williams, Jiahui Huang, Jonathan Swartz, Gergely Klar, Vijay Thakkar, Matthew Cong, Xuanchi Ren, **Ruilong Li**, et al.

- “Nerfstudio: A modular framework for neural radiance field development,” *ACM SIGGRAPH Conference Proceedings*, 2023, Matthew Tancik*, Ethan Weber*, Evonne Ng*, **Ruilong Li**, Brent Yi, Terrance Wang, Alexander Kristoffersen, et al.

- “NeRF-Det: Learning geometry-aware volumetric representation for multi-view 3D object detection,” *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, Chenfeng Xu, Bichen Wu, Ji Hou, Sam Tsai, **Ruilong Li**, Jialiang Wang, Wei Zhan, et al.

- “Nerfacc: Efficient sampling accelerates NeRFs,” *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, **Ruilong Li**, Hang Gao, Matthew Tancik, Angjoo Kanazawa.

- “Monocular dynamic view synthesis: A reality check,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2022, Hang Gao, **Ruilong Li**, Shubham Tulsiani, Bryan Russell, Angjoo Kanazawa.

- “TAVA: Template-free animatable volumetric actors,” *European Conference on Computer Vision (ECCV)*, 2022, **Ruilong Li**, Julian Tanke, Minh Vo, Michael Zollhöfer, Jürgen Gall, Angjoo Kanazawa, Christoph Lassner.

- “AI Choreographer: Music Conditioned 3D Dance Generation with AIST++,” *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, **Ruilong Li***, Shan Yang*, David A. Ross, Angjoo Kanazawa.

- “Plenoctrees for real-time rendering of neural radiance fields,” *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, Alex Yu, **Ruilong Li**, Matthew Tancik, Hao Li, Ren Ng, Angjoo Kanazawa.

- “Deep portrait image completion and extrapolation,” *IEEE Transactions on Image Processing (TIP)*, 2019, Xian Wu, **Ruilong Li**, Fang-Lue Zhang, Jian-Cheng Liu, Jue Wang, Ariel Shamir, Shi-Min Hu.

- “PortraitNet: Real-time portrait segmentation network for mobile device,” *Computers & Graphics*, 2019, Song-Hai Zhang, Xin Dong, Hui Li, **Ruilong Li**, and Yong-Liang Yang.

- “Pose2seg: Detection free human instance segmentation,” *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, Song-Hai Zhang, **Ruilong Li**, Xin Dong, Paul Rosin, Zixi Cai, Xi Han, Dingcheng Yang, Haozhi Huang, Shi-Min Hu.

- “Example-guided style-consistent image synthesis from semantic labeling,” *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, Miao Wang, Guo-Ye Yang, **Ruilong Li**, Run-Ze Liang, Song-Hai Zhang, Peter M. Hall, Shi-Min Hu.

- “Detecting and removing visual distractors for video aesthetic enhancement,” *IEEE Transactions on Multimedia (TMM)*, 2018, Fang-Lue Zhang, Xian Wu, **Ruilong Li**, Jue Wang, Zhao-Heng Zheng, Shi-Min Hu.

AWARDS & HONORS

BAIR Research Ignition Award, UC Berkeley **2021**

Best Show Award, ACM SIGGRAPH **2020**

– *“Volumetric human teleportation”, Siggraph Real-time Live!*

Best Demo Award, China Multimedia Conference **2017**

Kwang-Hua Scholarship, Tsinghua University **2014**

National Top 200 (0.04%), Chinese Physics Olympiad (CPhO) **2012**

INVITIED TALKS & INTERVIEWS

Interview / Podcast **Aug. 2024**

– *“Radiance Field and Gaussian Splatting”(link)*

EpicGames Research invited talk **Jun. 2023**

– *“Reconstructing Dynamic Objects from Video Captures via Neural Rendering”*

Tencent AI Lab invited talk **Dec. 2022**
– *"Acceleration Techniques and Toolbox for Neural Radiance Field"*

Adobe Research invited talk **Sep. 2021**
– *"Learning to Digitize Human Performance"*

MPI invited talk **Feb. 2021**
– *"AI Choreographer: Learn to dance with AIST++" (link)*

USC Viterbi Magazine interview **Jan. 2021**
– *"Connecting People in a Distanced World" (link)*

ACM SIGGRAPH interview **Oct. 2020**
– *"We're One Step Closer to Consumer-accessible Immersive Teleportation" (link)*